Thuan Doan

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Experience

Spartan — **Nov 2024 – Present**
Senior AI Engineer — *Ho Chi Minh, Vietnam*

- Built data pipelines to extract biomedical entities from clinical trials, PubMed, and FDA drug databases, constructing a knowledge graph to power advanced healthcare analytics and insights.
- Applied context engineering and advanced Retrieval-Augmented Generation (RAG) techniques, including GraphRAG and re-ranking models, to improve retrieval accuracy and contextual reasoning in chatbot and AI systems.
- Developed a medical deep research agents that integrates multi-source retrieval, biomedical tools, and reasoning capabilities to accelerate literature review, fact checking and clinical insights discovery.
- Optimized model deployment and monitoring, improving scalability, efficiency, and reliability of production systems.

HCLTech — **Jun 2023 – Oct 2024**
AI Engineer/Researcher — *Ho Chi Minh, Vietnam*

- Designed and implemented agentic LLM workflows to automate business processes, integrating internal tools, policies, and domain knowledge for higher accuracy and usability.
- Built data pipelines and fine-tuned Large Language Models (LLMs) for API test automation at ANZ Bank, improving accuracy and reducing manual effort.
- Developed and deployed GUI agents for web and mobile end-to-end testing, enhancing overall testing efficiency.
- Conducted research on advanced LLM techniques—including reinforcement learning, quantization, test-time scaling, and multi-agent systems—exploring novel approaches to improve adaptability, efficiency, and performance.

Infinigru — **Aug. 2018 – Apr 2023**
Machine Learning Engineer — *Korea/Vietnam*

- Developed and optimized machine learning models for fintech applications such as credit scoring, fraud detection.
- Built and fine-tuned vision models for object detection, segmentation, action recognition, and depth estimation.
- Designed, trained, and deployed NLP models for text classification, sentiment analysis, NER, and speech-to-text tasks.
- Implemented and maintained end-to-end ML pipelines, covering data preprocessing, feature engineering, model training, evaluation, and deployment across on-premise and cloud platforms.

Global Outsource Solution — **Mar 2015 – Mar 2018**
Technical Lead — *Ho Chi Minh, Vietnam*

- Designed and developed architecture of complex systems, ensuring scalability, performance, and maintainability.
- Led cross-functional teams of developers, designers, and QA, driving collaboration and on-time delivery.
- Applied CI/CD pipelines and software engineering best practices, improving code quality and accelerating deployments.
- Contributed hands-on coding and supported team members in solving complex technical problems.

Global CyberSoft/Forix — **Sep 2008 – Mar 2014**
Software Engineer — *Ho Chi Minh, Vietnam*

- Developed web applications across domains such as e-commerce, CMS, social media, and data warehousing.
- Built web and backend applications with frameworks such as Laravel, Zend, Magento, Wordpress, ASP.NET and optimized RESTful APIs and database schemas to ensure high performance and scalability.
- Enhanced front-end using HTML, CSS, and JavaScript, ensuring responsiveness and cross-browser compatibility.

Education

University of Information Technology — **Ho Chi Minh, Vietnam**
Master's Degree in Computer Science — *2020 – 2023*

- Research assistant at MMLab. Topics: NLP, LLMs, VLM, SLAM, 3D vision, deep generative models.

Japan Advanced Institute of Science and Technology (JAIST) — **Ishikawa, Japan**
Research Assistant under supervision of Prof. Kazuhiro Ogata — *Jan-Mar 2020*

- Research Topic: Theorem proving and state machine with CafeOBJ.

Kookmin University — **Seoul, Korea**
Research Assistant under supervision of Prof. Young Kim. — *2011 – 2012*

- Research Topic: Security-Enhanced Linux (SELinux) for web servers.

University of Science — **Ho Chi Minh, Vietnam**
Bachelor's Degree in Computer Science — *2004 – 2008*

Technical Skills

Languages: Python, PHP, JavaScript, Java.
ML Algorithms: Regression, Classification, Tree-based Models, CNN, RNN, Transformers.
AI Frameworks & Tools: TensorFlow, PyTorch, Scikit-learn, Hugging Face, Pandas, Numpy, Matplotlib, OpenCV, NLTK, MLflow, Datadog, Sentry, Prometheus, Grafana, SageMaker, VertexAI.
LLM Tools: LangChain, Google ADK, LlamaIndex, Langfuse, LiteLLM, vLLM, Ollama, Unsloth, DeepEval, Guardrails.
App Frameworks: FastAPI, Flask, Spring, Laravel, React, NodeJS, Angular, VueJS, Magento, WordPress.
Data: MySQL, PostgreSQL, MongoDB, Redis, Neo4j, Apache Airflow, Kafka, Flink, PySpark, ChromaDB.
Others: Git, Docker, Kubernetes, AWS, GCP, Arduino, Raspberry.

Achievements

Certificates

- Tensorflow Developer - Certificate
- Coursera specializations about Math, Machine Learning, Deep Learning, Vision, NLP, Data Science - Certificates
- IELTS 6.5 - Certificate

Publication:

- Efficient Finetuning Large Language Models For Vietnamese Chatbot (first author) MAPR 2023

Projects

TriangleHealth (2025) | *Apache Airflow, TxGemma, GraphRAG, Neo4j, Deep Research Agents, Google ADK, MCP, vLLM.*

- Built data pipelines with Apache Airflow to extract biomedical entities and construct a knowledge graph, while processing and indexing patient healthcare records using SmolDocling, BiomedBERT, and pgvector.
- Applied advanced Retrieval-Augmented Generation (RAG) techniques—including GraphRAG with Neo4j, re-ranking with TourRank, and context engineering—to enhance retrieval accuracy and contextual reasoning in AI chatbots.
- Developed a medical research agent using Google ADK, integrated with biomedical models (TxGemma, Biomni) to autonomously perform literature review, fact-checking, and clinical research tasks.
- Optimized model deployment using vLLM and exposed capabilities via MCP servers.
- Implemented tracing and observability with Langfuse and Datadog.

EnsayoAI (2023-2024) | *Llama3, LangGraph, RAG, langfuse, promptfoo, deepeval, pgvector, FastAPI, AWS SageMaker*

- Developed a generative AI testing automation tool using open-source LLMs (Mistral, Llama 3) with domain-adaptive fine-tuning techniques including LoRA and RLHF to optimize performance for software testing workflows.
- Built an intelligent chatbot system enhanced with advanced RAG architecture featuring hybrid search, semantic routing, optimized chunking strategies, re-ranking algorithms, multi-retriever, multimodal capabilities, and tool-integrated.
- Architected and implemented a multi-agent LLM framework using LangChain and LangGraph to automate complex business workflows within banking operations, enabling autonomous task execution.
- Designed and deployed GUI automation agents for comprehensive web and mobile end-to-end testing.
- Implemented LLM evaluation pipelines with DeepEval and Promptfoo to assess accuracy and robustness.

GruBot (2022) | *Python, wav2vec, Kaldi, RASA, nltk, BERT, LSTM, Kubernetes, MLflow, Prometheus, Grafana.*

- Developed and fine-tuned Korean and Vietnamese speech recognition models using Kaldi and wav2vec.
- Built RASA-based conversational AI chatbot with NLU pipeline and dialogue management, handling multi-turn conversations, custom actions via API integrations, and seamless human agent hand-off for complex queries.
- Optimized BERT-based models for Named Entity Recognition (NER) and Sentiment Analysis.

GruFDS (2021) | *Python, Java, Spring Boot, Apache Kafka, Apache Flink, PySpark, XGBoost, LightGBM, SHAP, VertexAI*

- Developed high-performance real-time fraud detection system using hybrid approach combining rule-based systems with ML models (Random Forest, XGBoost, LightGBM) to reduce false positives and detect evolving fraud patterns.
- Built real-time data pipeline using Apache Kafka and Flink for processing payment transactions, implementing threshold-based decision-making to classify transactions as fraudulent, suspicious, or normal

AirPatrol (2021) | *U-Net, DeepLab, Mask R-CNN, TensorFlow, Xception, Jetson Nano*

- Developed automated crack detection and wildfire classification systems using deep learning models (U-Net, DeepLab, Mask R-CNN, Xception) for infrastructure and UAV surveillance.
- Optimized and deployed TensorFlow Lite models for real-time inference on resource-constrained UAV platforms.